LATYS

Networks of the Future



FOCUS V1.0 USER GUIDE

Prepared for Western University

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FOCUS V1.0 USER GUIDE

- This document provides a step-by-step guide to use LATYS FOCUS V1.0 to eliminate dead spots in an existing network, or to extend to where there was previously no coverage.
- The device is designed for 5.85GHz and the enclosure is designed for indoor use.
- Please send your test results or questions to research-development@latys.ca.

IN THE BOX

LATYS FOCUS

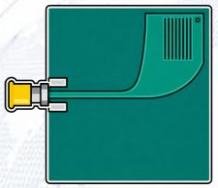
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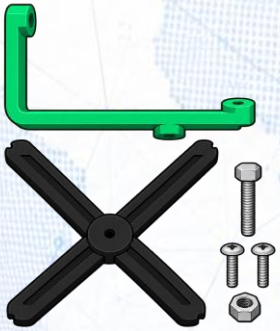
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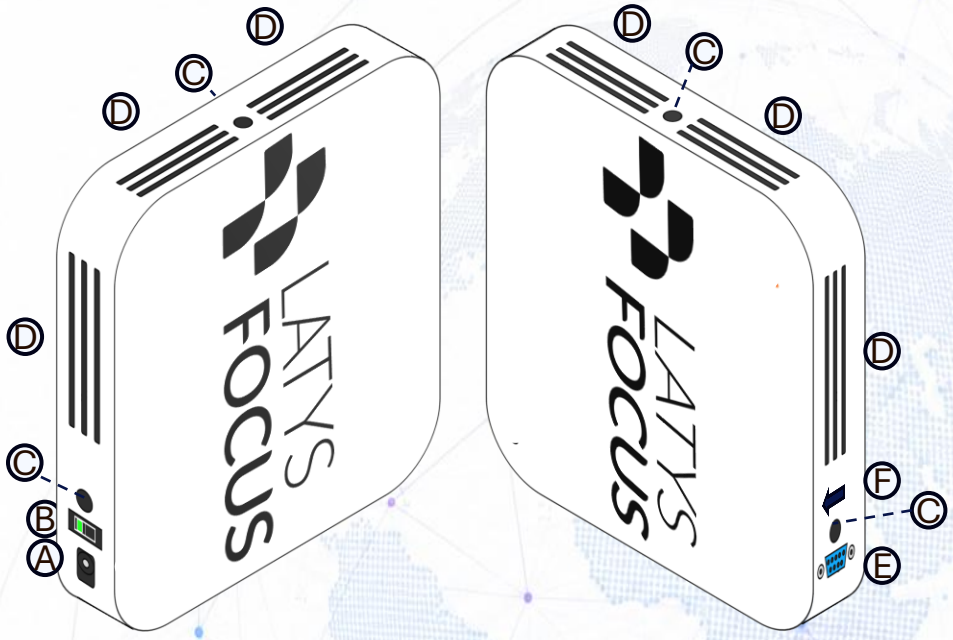
LATYS FOCUS

LATYS Focus is a Signal Amplifier which enhances data transfer by increasing the power of a signal. The unit is designed to improve coverage in dead zones.

The device is designed to receive and transmit the RF spectrum in 5.85GHz.

OVERVIEW

A. Power Plug
B. Power Switch
C. Mounting Hole
D. Vents
E. Control Port
F. Arrow Indicating Direction of Strong Outgoing signal



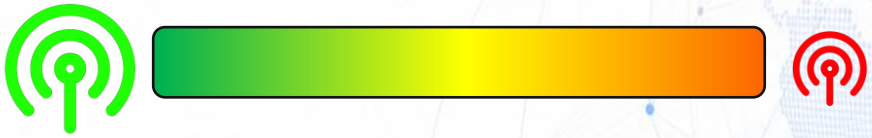
HOW TO START

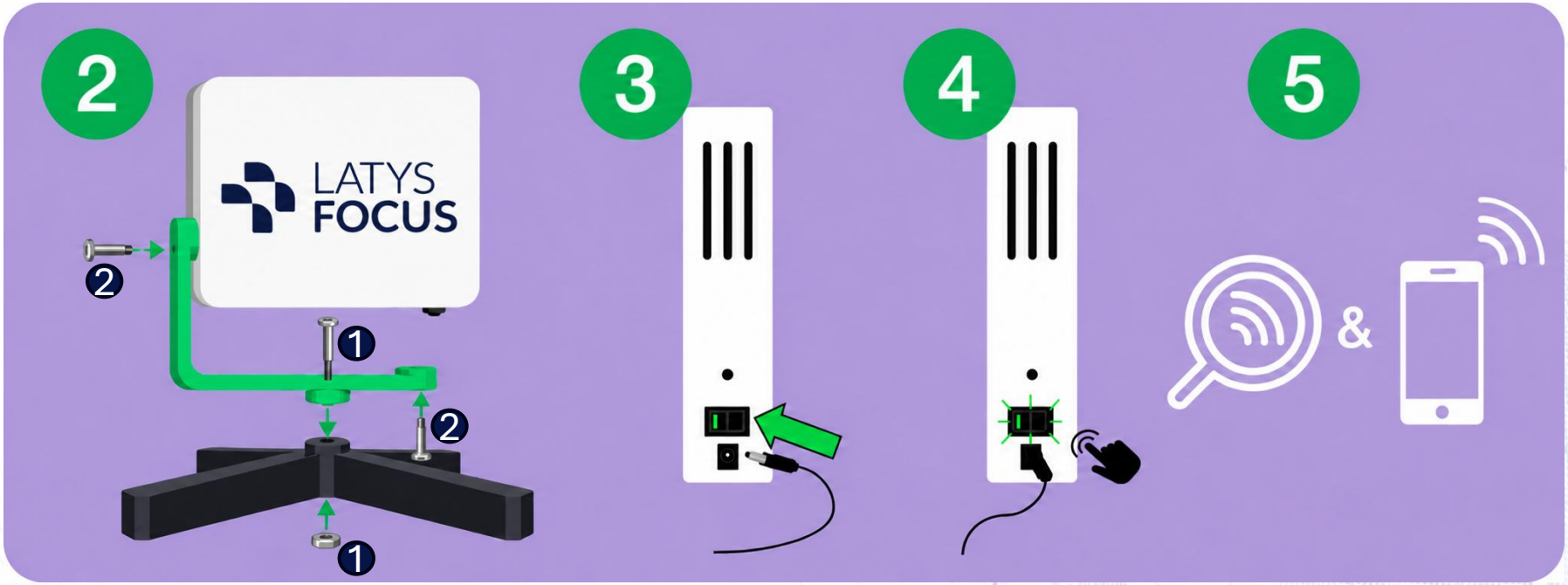
1. Choose the area where you want to improve signal coverage and identify where to place the panel.
   *For best performance, the panel should be positioned 50 cm to 150 cm from the transmitter, with 100 cm as the recommended distance. The arrow on the side of the panel indicates the direction of the signal.
2. Position the panel based on your test setup.
   The panel can be mounted in different ways. Use a wall holder to mount it on a wall, place the unit on a flat, stable surface, or use the mounting holes to attach it to a standard tripod or another suitable mounting system.
3. Connect the power.
   Plug the LATYS FOCUS into a power outlet using the provided power adapter.
4. Turn on the LATYS FOCUS.
5. Check the signal strength in the target area.

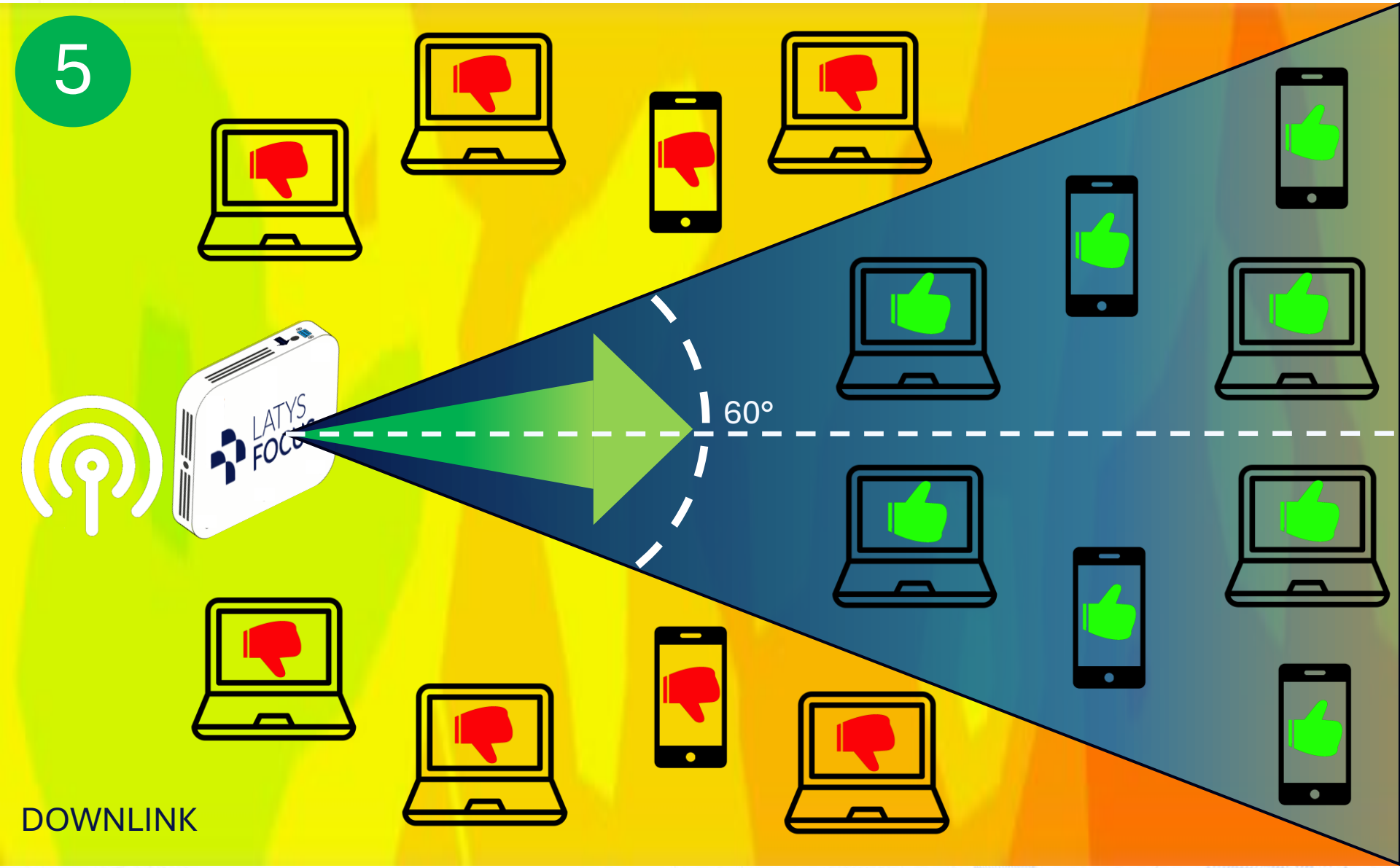
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* Optimal distance is 1m

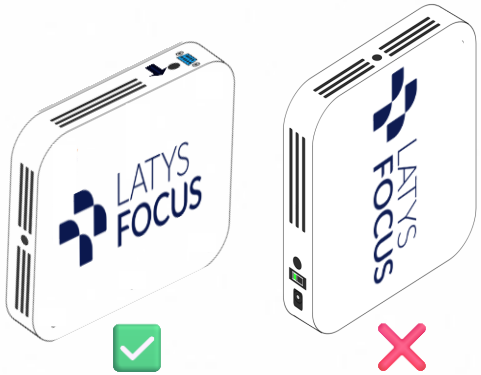



2
LATYS
FOCUS
2
1
2
1
3
4
5
&


5
60°
DOWNLINK

BEAM STEERING

1. Check the device orientation:
   To apply the beam steering in azimuth plane , make sure the device is oriented correctly:



2. Connect the beam controller to the LATYS FOCUS panel.
3. Adjust the angle and apply the configuration:
   a. By using the control buttons
   b. By using a Computer



Beam Controller

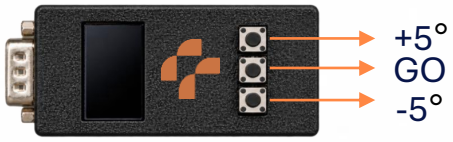


*Maximum performance is obtained at **5.8GHz**

ADJUSTING THE ANGLE

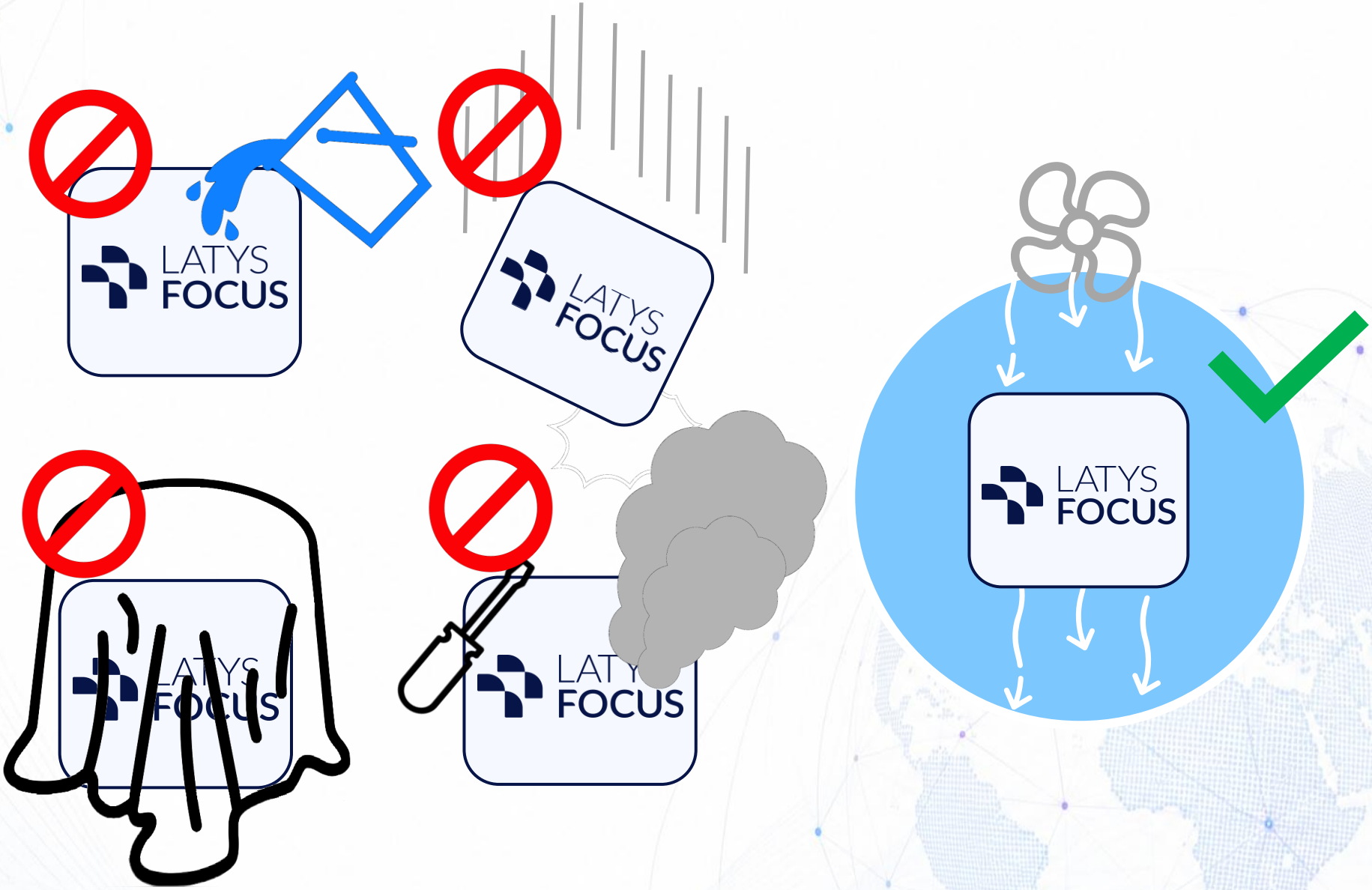
a. Using the control buttons:

- Press + to increase the angle by 5°.
- Press − to decrease the angle by 5°.
- Press GO to apply the selected angle



b. Using a computer to change the configuration

- Connect the beam controller to LATYS FOCUS and your PC.
- Open Arduino software on your PC (or you can use any other USB-to-serial Software)
- Open a new sketch by pressing Ctrl+N.
- Copy and paste the Arduino script provided into the new sketch.
- Click the Serial Monitor icon at the top-right of the Arduino software.
- Set the baud rate to 115200.
- Set the line ending option to Both NL & CR.
- Click the Verify icon to check the code.
- Click the Upload icon to upload the code to the microcontroller.
- After uploading, use the Serial Monitor message bar to enter the desired steering angle.
- Enter an angle between −35° and +35° in steps of 5°, then press Enter


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FOCUS

IMPORTANT

- Be sure to insert the power plug connector appropriately into the unit. Do not apply excessive force to avoid breaking the device.
- Do not use any other accessories other than those provided or explicitly recommended by LATYS.
- LATYS FOCUS improves the signal in a 60° cone pattern. Devices located inside this cone can observe improvement of the signal.
- The device is meant for indoor use.
  Operating temperature range is +5° to +35°C (+41°F to +95°F).
- The device will get warm during use. This is normal and expected. Avoid touching the warm parts unnecessarily.
- LATYS FOCUS does not have an active radio, so no MAC address is associated to the device.

CAUTION

- It is not permitted to use the device for anything other than what it was originally intended for, as described earlier.
- Do NOT immerse the device or the power supply in water, even partially. Failure to follow this direction risks serious injury or death.
- Do NOT try to open or tamper with the device mechanically in any way. Failure to follow this directive risks serious injury or death.
- Do NOT attempt to modify the product. The LATYS FOCUS is not designed to be serviced by the user. Contact LATYS support when necessary.
- Do NOT expose the device to excessive moisture, fluids or liquids, heat, fire, smoke, or corrosive substances.
- Do NOT use or store the product in a location where the temperature exceeds 35°C / 95°F.
- Use only the power supply provided by LATYS. Never exceed the maximum allowed input rating of the power supply.
- Do NOT use the device when it is covered or placed inside an enclosed space e.g. a box or a bag.
- Do NOT drop the device or strike it with hard objects.
- Do NOT let children or pets play with the device.

THANK YOU

RE-IMAGINING A MORE CONNECTED FUTURE

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